Committee: World Health Organization (WHO)

Topic: The Question of Strengthening Chronic Disease Management through the Global Standardization of Healthcare Systems

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Introduction

While globalization has generated extraordinary benefits to humanity, this has also resulted in various health risks regarding chronic disease to disseminate rapidly across the world. Nevertheless, based on a study that was published in The Lancet about “A Systematic Analysis for the Global Burden of Disease”, between 2010 and 2023, major leading causes of premature death and healthy life lost to illness or disabilities, such as risks that are related to child and maternal health and contaminated environments, have declined. The diseases that have instead surged are mostly those that are related to chronic diseases, which are also known as noncommunicable diseases (NCDs). Chronic illnesses can cause a wide range of symptoms, and some examples include chronic pain, physical limitations such as lower muscle strength, fatigue, body changes, and even mental health effects. Chronic illness causes disruption to their lives; a few examples include time demands, financial strain, and loss of hobbies because of it. Simon Hay, the director of research strategy at the Institute for Health Metrics and Evaluation (IHME), once stated that "NCDs are by far the biggest thing that kills human beings on this planet, and they're getting worse.” These kinds of statements show that currently chronic disease is a massive problem for our global society. According to the World Health Organization’s Fact Sheet, chronic disease killed at least 43 million people in 2021, which is around 74% of non-pandemic-related deaths globally. In addition to this, according to the OECD, chronic conditions such as cancer and diabetes are the current leading causes of death in the OECD countries, as well as causing significant disabilities. These diseases tend to have a long duration, and a combination of genetic, physiological, environmental, and behavioral factors are mainly the cause of them. The 2030 Agenda for Sustainable Development has even acknowledged those as a major obstacle to sustainable development. By 2030, it aims to reduce one-third of mortality from chronic diseases with the use of prevention and treatment, as stated in SDGs target 3.4.

Despite the clarity of these statistics, global response mechanisms are currently deeply fragmented. The management of chronic diseases requires sustainable, long-lasting tracking technologies that include a seamless connection across multiple networks instead of traditional ones that are against diseases that are mostly short-lived. Currently, patients with chronic conditions face harsh systemic barriers due to these medical systems lacking unification, while also having different data format variations across different nations' systems. This hinders the exchange of information and coordination between different health systems. In short, when a patient moves across nations, their medical history often gets lost in the process. For instance, over 1.2 million people in the USA have gone into Mexico for healthcare in 2024 alone, according to the American University. However, their administrative information is most likely to be lost because of the border crossing.

While the need for a unified response to these chronic diseases is apparent, the global standardization of healthcare systems has major systemic issues. Historically, WHO utilized the "4x4 framework", which focused on 4 major diseases and 4 behavioral risk factors, however, this standardization has been criticized for neglecting the poor. The 4x4 framework mostly prioritizes lifestyle diseases that were prevalent in MEDCs, and because of this, common diseases in LEDCs such as rheumatic heart disease and malnutrition-related (J-type) diabetes were disregarded. In addition to this, noncommunicable diseases receive a disproportionately small share of international funding. Noncommunicable diseases currently receive only 1~2% of all Official Development Assistance (ODA) for health, according to a study performed by Leah N. Schwartz. This study also states that in 2010, the investment for infectious diseases such as HIV and malaria was around 30 times higher than NCDs. Without a standardized global commitment to redirect financial resources, poorer nations will be unable to establish necessary healthcare infrastructures to combat chronic diseases. Moreover, effective chronic disease management requires health data to be transferred across the nation. However, technical interoperability standards often clash with national data sovereignty. Some nations have already implemented various regulations that prohibit the patient's data from being processed outside their territory. Yet, although facilitating the free flow of data is important for patient safety and global standardization of systems, it should also respect the nation's legal frameworks regarding data security and ownership. Therefore, global standardization should integrate standardization regarding data interoperability and standardized clinical guidelines to ensure smooth and equal chronic healthcare for all.

Definition of Key Terms

Chronic Diseases

A disease that continues for a long time, typically lasting one year or more, that requires ongoing medical attention and limits daily activities. The main examples of chronic diseases are cancers, diabetes, cardiovascular diseases, which include heart attacks and stroke, and chronic respiratory diseases, which are diseases such as chronic obstructive pulmonary disease and asthma. Usually, chronic diseases affect people in the low- and middle-income class. Statistically, from these classes, more than three-quarters of all global chronic disease deaths occur. Chronic disease is also known as non-communicable diseases (NCDs), however, they are not the exact same disease, although they share various traits.

Non-Communicable Diseases (NCDs)

A disease that is not passed from one person to another, meaning that the disease is not contagious. They cannot be caught by other people, like bacteria or viruses, but instead are caused by a combination of genetics, environment, lifestyle, and more. Examples of Non-communicable diseases are cardiovascular diseases, type 2 diabetes, cancer, and chronic respiratory diseases. Most major diseases are both chronic and non-communicable; however, not all non-communicable diseases are chronic.

Cardiovascular Diseases (CVDs)

A condition that affects the heart or blood vessels. Currently, according to the World Health Organization, cardiovascular diseases are the leading cause of death worldwide, accounting for nearly 32% of all global deaths, which is around an estimated 19.8 million people in 2022. In addition to this, approximately 80% of the world's deaths from cardiovascular diseases occur in low- and middle-income countries. The types of cardiovascular diseases include coronary heart disease(affecting blood vessels of the heart), cerebrovascular disease(affecting blood vessels of the brain), peripheral arterial disease(affecting blood vessels of the arms and legs), congenital heart disease(birth defects that affect the normal structure and development of the heart), and more. The risk

factors for cardiovascular diseases include unhealthy diet, physical inactivity, smoking, and alcohol. Common symptoms include heart attacks and strokes.

Chronic Respiratory Diseases (CRDs)

A disease that affects the airways and other structures of the lungs. The common types of chronic respiratory diseases include chronic obstructive pulmonary disease (COPD), asthma, occupational lung diseases, and pulmonary hypertension. Tobacco use is the main risk factor of it, but other additional factors include air pollution, chemicals, and dust. Although CRDs are not curable, various treatments that help open the air passageway help with shortness of breath and other symptoms.

Interoperability

The ability to work together with other systems or pieces of equipment such as devices and software to communicate. Interoperability enables technology that originated from various locations to work together without requiring special effort from the users.

Digital Health

The integration of information and communication technologies towards healthcare in order to improve how we manage illnesses, deliver medical care, and more. By combining technologies like data and software, it makes healthcare more accessible and personalized for many patients.

Global Susceptibility

Worldwide vulnerability or likelihood of a population or system being harmed or affected by a specific risk or condition. In public health and medicine, this term refers to the global population's lack of resistance or their susceptibility to diseases and pandemics. For instance, it may combine environmental factors like air quality or social determinants of health to determine which populations worldwide are most at risk for developing certain noncommunicable or infectious diseases.

Cross-border Healthcare

The provision of healthcare that is outside of their original country for partnered countries' people, often facilitated by international agreements between nations. For example, during the COVID pandemic in 2020, the European Union (EU) used this system to maximize utilization of resources. To share the burden of the COVID pandemic, countries that had capacity for patients accepted patients in critical condition, and borders were open for healthcare workers and people seeking medical care.

Clinical Practice Guidelines

Systematically developed guidelines that are designed to assist healthcare professionals and patients in making decisions about appropriate healthcare for them. By evaluating the results of various medical interventions, these frameworks improve patient outcomes and standardize the quality of care. However, they are used mostly as a flexible decision-making tool rather than a legally binding protocol.

Electronic Health Records (EHR)

An electronic version of a patient's medical history, which is maintained by the provider through the years. Electronic health records may include key administrative health data regarding that person. This automates access to information and has the potential to make the work more efficient. This information will enable the provider to provide better care and make healthier decisions.

Universal Health Coverage (UHC)

A service that allows all individuals and communities to receive the quality health services they need. This includes promotion, prevention, treatment, rehabilitation, and palliation. UHC brings together service coverage with financial protection in order to prevent medical expenses from pushing households into poverty. The coverage of UHC has stagnated in recent years, leaving approximately half of the world's population without essential health services. In addition to this, 2 billion people are facing harsh poverty because of healthcare spending.

Fast Healthcare Interoperability Resources (FHIR)

A free healthcare data standard created by Health Level Seven International in 2011, which allowed different EHRs and medical apps to share information safely and quickly. By enabling seamless data sharing across disparate platforms, FHIR is a basis for national and global health data interoperability.

International Classification of Diseases, 11th Revision (ICD-11)

WHO's digital first global standard for recording and reporting health conditions and mortality data. This officially went into effect in January 2022 after being adopted by the World Health Assembly in May 2019. ICD-11 ensures interoperability so that clinical diagnosis carries the same definition and meaning across different health platforms.

Official Development Assistance (ODA)

Government aid given to developing countries mainly to promote economic development and welfare, including aid spent on health. ODA's purpose is to target the economic development and welfare of countries listed on the OECD Development Assistance Committee. ODA excludes military aid, counter-terrorism operations, and commercial transactions. In regards to chronic disease management, noncommunicable diseases cause roughly 74% of all deaths worldwide, yet receive only 1–2% of health-related international aid.

Primary Health Care (PHC)

An integrated approach to health that brings essential services, from promotion and prevention to treatment and rehabilitation that is as close as possible to people's everyday environments. As the foundation for Universal Health Coverage, PHC is one of the most effective models for managing chronic illnesses early before they escalate into high-cost hospital emergencies.

Disability-Adjusted Life Year (DALY)

A measure of disease burden. One DALY equals one year of healthy life lost to early death or illness. "Cost per DALY averted" compares how much a program spends to prevent that loss.

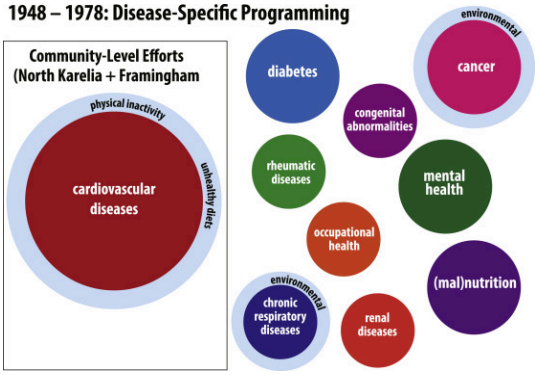
Background Information

History of Chronic Diseases

After the World Health Organization was established in 1948, the WHO's early reports anticipated that as developing nations undergo industrialization and urbanization, their diet and lifestyle habits would shift, which would eventually lead to the chronic disease patterns that were seen in industrialized nations at that time. The initial interest in chronic diseases first appeared from the WHO European Regional Office (EURO). During the first session of the Consultative Committee for Europe in May 1951, the Committee recommended the creation of specific WHO programs targeting "non-infectious chronic diseases of adult populations". Despite this early interest in chronic diseases, the activities were mostly operated independently for each disease, meaning that the diseases did not have any unifying framework at that time.

The epistemological frameworks developed by CVD epidemiologists can be seen from the WHO as early as from 1955 in the Recommendations of the Fifth Session of the Regional Committee of Europe. At the 1957 Symposium on the Public Health Aspects of Chronic Disease in Amsterdam, global health officials grouped four "degenerative diseases", which were CVD, diabetes, respiratory illnesses, and cancer, according to the cause explanations of specific risks, behaviors, and lifestyles.

Framingham Heart Study, done by the United States National Heart Institute and Public Health Service (NHI) and launched in 1948, created the notion that chronic diseases can be treated by a set of shared behaviors. This study showed a remarkable move during that time by deploying the tools and techniques that were used in epidemiology, historically utilized in the study of infectious diseases, to analyze the causes of chronic diseases and non-communicable diseases. It successfully identified lifestyle risk factors and their strong statistical correlation with the development of cardiovascular diseases. The Framingham Heart study was used in various templates of modern nations, one of the examples being the North Karelia Project, which was done in Finland and was launched in 1978. Based on the intellectual framework of NCD risk factor analysis, this project extended it with the aim of developing community-based interventions to decrease the communities' lifestyle risk factors. One of the results of this study was that further encouraging general changes in unhealthy lifestyles, especially during the era of economic transitions, could possibly result in the mitigation of NCD rates.



Disease-specific programming at the WHO and the emergence of community-level epidemiography characterization of CVD (1948–1978) - Sciencedirect

After Framingham and North Karelia, WHO tried to strengthen the programs that were related to NCDs and chronic diseases by utilizing a more integrated approach. In the early 1980s, WHO found out that the risk factors that were associated with CVDs, such as unhealthy diet and the use of tobacco, were also connected with other diseases like cancer, diabetes, and respiratory diseases. In 1982, the WHO expert committee concluded that efforts that tried to prevent coronary heart disease should be a more general program against NCDs. This led to the launch of the Countrywide Integrated Noncommunicable Diseases Intervention (CINDI) program in 1984. CINDI went for an integrated approach to NCDs, rather than the original small-scale projects in the past. In 1990, after the CINDI program, the INTERHEALTH program was launched with the aim to integrate community-based approaches to all WHO regions. This meant that the "Western" type of NCD management was effectively standardized for the rest of the world. In May 1998, Resolution WHA51.18 in WHO officially started the proposed framework for an integrated approach to preventing and managing chronic noncommunicable diseases.

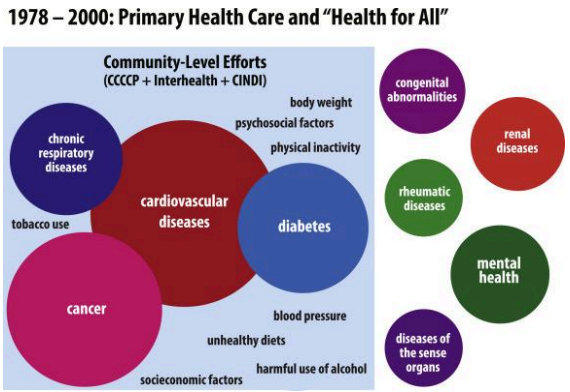


Illustration of the emergence of shared risk factors as categorizing schema for NCDs (1978-2000) - Sciencedirect

The “4x4 Framework” was established after 2000. In May 2000, the 53rd World Health Assembly passed resolution WHA53.17, adopting the "Global Strategy for the Prevention and Control of Noncommunicable Diseases". This standardized by maintaining the focus on 4 sets of diseases and 4 shared behavioral risk factors. The diseases were CVD, cancer, diabetes, and chronic respiratory diseases, and the risk factors were tobacco use, harmful use of alcohol, unhealthy diets, and physical inactivity. This framework became a simplified governance tool, and it also allowed the WHO to provide straightforward recommendations, as well as standardized monitoring tools worldwide. This framework was further developed in 2008, when the WHO released the Action Plan for the Global Strategy, which prominently featured the 4x4 model.

In the last decade, the international community has acknowledged the fact that there are limitations to the original 4x4 system, starting with the UN High-Level Meeting on NCDs in 2011. By the Third UN High-Level Meeting in 2018, the global framework officially shifted into a “5x5 agenda”. This up-to-date standardization acknowledged the critical roles of mental health, and environmental risk factors. This includes factors such as air pollution. Recently, the focus of the WHO and the global society have shifted into a more “precision public health”, meaning that it is balancing the healthcare system between global standards and people-centered care.

Risk Factors of Chronic Diseases

Various factors, especially those related to lifestyles, could cause chronic diseases. Cigarette smoking, for example, causes more than four hundred and eighty thousand deaths each year in the US. In addition, over 16 million people are living with a disease caused by smoking in the US alone. Smoking causes various chronic diseases such as cancer, heart disease, stroke, lung diseases, diabetes, and chronic obstructive pulmonary disease (COPD). COPD is a progressive lung condition that makes it difficult to breathe, and causes various symptoms such as shortness of breath, chest tightness and more to develop over time, and is the one of the leading causes of death in the United States. Other causes of it may include smoking, poor nutrition and physical inactivity, and excessive alcohol use.

In addition to this, an unhealthy diet, which is mostly known as taking excessive amounts of salt, sugar, and saturated fat, and insufficient physical activity are also significant risk factors for chronic diseases. These factors are the main cause of obesity and other various chronic diseases such as type 2 diabetes, heart disease, some cancers, stroke, and even depression. Although these behaviors can be caused by individual choices, the environment also affects this factor considerably. An excessive amount of alcohol can result in serious problems, which may include alcohol use disorder

and problems regarding memory and learning. These could result in chronic health conditions like high blood pressure, heart disease, stroke, and some cancer.

Often, these behavioral risks result in metabolic risk factors, which show the current progress for severe health conditions. The most eminent one is hypertension, which is having a high blood pressure. Other risks include being overweight, obesity, high blood glucose, and more. As nations get economically developed and move into a more modern styled lifestyle, these metabolic risks will arise.

Even though it may seem like that only behavioral and metabolic risk factors that are shown above exist, in fact, other risk factors regarding points such as environmental and developmental determinants occur. Some groups have a more significant chance of developing chronic diseases because of their background conditions. This may include where they are born, live, work and age. These non-medical factors, which are called social determinants of health, can either affect the risk of chronic diseases in a positive or negative way. When negative factors increase, they limit the opportunities to make healthy choices and get good medical care. For instance, some places have a shortage in places like parks for people to be active, or healthy stores that sell fresh vegetables and fruits. In some rural areas, it is hard to get medical care because they may lack doctors in their areas, or they have to travel long distances to reach the nearest hospitals. This makes it challenging for them to get preventive screenings or specialist follow-up care.

Difference of Chronic Disease Aspect in Low, Middle, and High Income Nations

Chronic disease is a common disease worldwide. Yet, according to WHO, around 80% of all deaths caused by chronic disease are concentrated in areas that have low or middle income. As these nations undergo rapid urbanization and industrialization, LEDCs would face chronic diseases because of the change in lifestyle, as well as remaining infectious diseases. As the detection for these diseases is mostly late in the course of the disease because of lack of infrastructure, people die at a younger age from chronic diseases, NCDs, and CVDs, mostly in their productive working years. According to WHO, in 2021, around 82% of NCD deaths that occurred before the age of 70 took place in low and middle income nations. This is because people living in low and middle income countries mostly do not have primary healthcare welfare programs and infrastructures with the purpose of early detection and treatment for patients that have risk factors for these diseases. In contrast, in MEDCs, standardized testing protocols enable the early management of risk factors like hypertension and more.

Furthermore, the economic impact of chronic diseases for people with lower income in LEDCs is critical. Treatment of chronic disease and NCDs require catastrophic health spendings, and this high cost can push vulnerable populations to extreme poverty, or even die because they have limited access to treatments. Despite this, a massive financing disparity exists in the global health aid. According to a study performed by Leah N. Schwartz, noncommunicable diseases currently receive around 1~2% of all Official Development Assistance (ODA) for health, even though NCDs cause around 74% of all deaths worldwide. Yet, historically, infectious diseases like HIV and malaria have historically been over 30 times higher. According to research by Nugent and Feigl in 2010, international donors invested only $0.78 per disability-adjusted life year (DALY) attributable to NCDs. But, in HIV, malaria care, treatment, and prevention programs, a better yet inadequate amount of $23.90 per DALY was invested in developing nations. This highlights a massive global funding disparity. In addition to this, the “4x4 program” mentioned above, although simple, has a downside of focusing only on modern lifestyles such as tobacco and dietaries. Chronic conditions that primarily affect the poor, such as rheumatic heart disease and malnutrition-related diabetes were ignored in comparison. For the goal of global standardization to be successful, these issues should be also addressed.

Problem of Fragmentation of Systems

To improve global chronic disease care, the lack of standardized systems as well as fragmentation in the systems worldwide should be solved. Chronic conditions require long-term and sustainable tracking systems, but current response mechanisms are separated from one another. Interoperability is especially critical to solve this problem. Currently, however, medical records frequently exist in “silos”, meaning that when the patient gets treatment on both sides of the border, the data exists separately. In many regions, even in this modernized world with digital health tools, it may result in stand-alone solutions that result in poor service because of the fact that whenever they switch providers or facilities, their medical data is fragmented. There are various examples of these "healthcare ghosts", which are terms that refer to patients who lose critical pieces of their medical history every time they cross a national border. Along the US-Mexico border, over 1.2 million Americans crossed into Mexico for healthcare in 2024 alone. The regulations in the United States’ HIPAA do not apply to information obtained outside its borders, and this creates a fragmented system where patients must manually piece together their own healthcare narratives. This fragmentation means that in addition to treatments for the patients, a scientist in a country could not access the treatment notes or lab results from a country in another nation. Moreover, some US States

implemented data sovereignty laws to protect citizen privacy, such as blocking the access of data outside of their territory, but these often act as unintended barriers to care systems.

Global Standardization of Healthcare

Care standardization is the process of creating and implementing evidence-based practices consistently across healthcare providers to ensure efficient, high-quality, and fair patient care. These predefined approaches shape the best practices for the screening, treatment, and long-term monitoring of specific chronic conditions. The benefits of it include better patient outcomes, lower clinical disparity, less medical inaccuracies, streamlined workflows, and less overall healthcare fees.

For global standards to be effective, systems must achieve interoperability. This is achieved through technical interoperability and semantic interoperability. First, for technical interoperability, the Fast Healthcare Interoperability Resources (FHIR) standard has emerged as the global benchmark for the exchange of electronic health records (EHRs). It allows data to move seamlessly between authorized providers and thus ensure the patient's medical narrative to be continuous. To ensure that clinical data is understood correctly across languages and systems, the WHO's International Classification of Diseases (ICD-11) provides a common medical language. This ensures that a diagnosis or treatment coded in one Member State has the same clinical meaning in another one. Historically, chronic disease management was often concentrated on isolated and disease-specific programs. However, after the modern globalization of the world, WHO urged all nations to move toward a more integrated and people-centered health system. With this approach, it could create a continuous care system by standardizing how information is captured and shared across different levels of healthcare, such as primary community clinics and specialized hospitals. This places the patient, rather than the disease, at the center of this healthcare system. To sum up, standardization of healthcare system includes data standards like common formats and vocabularies so records can travel with the patient such as HL7 FHIR and ICD-11, as well as shared and evidence-based clinical guidelines for treatment, which includes WHO HEARTS technical package for hypertension and CVD in primary care, and shared principles on primary health care, UHC, and how domestic and international money is allocated.

Various Current Global Efforts

Most importantly, the 2030 Agenda for Sustainable Development acknowledges that NCDs are a primary driver of poverty and a barrier to economic growth. In addition to this, under the

SDGsTarget 3.4, all nations aim to reduce premature mortality from NCDs by one-third by 2030 through prevention and treatment. The World Health Assembly (WHA) also called for the Implementation Roadmap 2023-2030 to support actions across nine voluntary global targets. The WHO Global Strategy on Digital Health emphasizes the development of global minimum specifications for electronic health records (EHRs) and the promotion of digital public goods, which are open-source software and standards that can be shared across borders that also respects nation's sovereignty.

Recent Use of Advanced Intelligence (AI) and Technology

Digital transformation, especially the use of Artificial Intelligence (AI) and Big Data, is used variously for chronic disease management. AI-driven Clinical Decision Support Services (CDSS) tools are used to process complex patient data to provide doctors with real-time and evidence-based advice. These systems could assist in accurate diagnosis and reduce medical errors. For instance, modern platforms can automatically calculate cardiovascular risk scores and suggest personalized treatment goals or medication changes tailored to each patient. In addition to this, the integration of fragmented data from hospitals and pharmacies enables big data analytics. By using unique identifiers to link each separate record, health systems could perform real-time observation of a patient and create alerts for high-risk populations. Utilizing Internet of Things (IoT) and smart wearables, as well as portable testing equipment, all with standardized clinical data management capabilities also allows accurate, standardized, and real-time management of various health data of the patient such as blood pressure and blood glucose. With this, data collection is available even when the patient is outside hospital or other medical facilities.

Possible solutions

Although there are various obstacles in the way of solving this problem, nations can utilize various methods and cooperate together in order to solve the current problems. Because conditions differ greatly between countries, solutions should mostly be adaptable to each. Every country, regardless of income, should take part in standard-setting and in implementation. The development of various software applications that adhere to the global standards to specifically combat the challenges of chronic disease management is the key to solving this problem, and for this, while MEDCs can focus on technological innovation, LEDCs should mainly establish local adaptation and feasibility assessments in order to ensure global standards to be flexible and practical for various environments without creating a 'one-size-fits-all' approach, which is an approach that is applied to everyone or every situation without any modification. It is also worthwhile to note that as WHO is a normative organization, its resolutions and guidelines are voluntary and non-binding.

Strengthening Data Interoperability

To ensure the secure exchange of health information, nations should develop a national digital health infrastructure roadmap or guideline, adopt open-source health software, and aim for sustainable systems or assets that have interoperability of health information systems at national or even international levels. This can be done in various ways, and a few examples include establishing a national agency for creating national benchmarks, or interoperability strategies for exchange of secure data. Nations may also establish national patient data systems that serve as information hubs which are managed by patients themselves. This makes continuity of care possible even when patients move across borders by allowing patients to record these data themselves. To respect sovereignty, data should stay under national legal control and move only by agreement. Having common minimum privacy and consent principles could make national laws compatible without replacing them.

Developing Technical Standardization

The modern data-based healthcare system involves technical standardization. Some member states already have legal frameworks that make care providers use software that conforms to national standards, and software developers to develop the software that fulfills global guidelines. Some global standards include HL7 FHIR for electronic messaging, SNOMED-CT for comprehensive clinical health terminology and standardized codes for EHR, LOINC for universal codes for lab tests and

clinical observations and ICD-11 for classification of diseases and health conditions. Developing nations should also create comprehensive frameworks for the full implementation of health data standards that focus on creating the required technical capacity and resources to maintain these systems. WHO can publish global minimum specifications as a baseline, and countries then could adopt them in stages. These technical standardizations should be voluntary technical support and monitoring and not enforcement.

Initiating Digital Public Goods

The creation of digital public goods, which could be a software tool or a data model that could be shared nationally or globally, is critical for developing nations because they currently have a huge technological gap between MEDCs and LEDCs. By creating this digital model, LEDCs could use MEDCs technology to solve the chronic disease issue. Other solutions may involve the investment and deployment of these technologies that are easily adaptable to low-resource settings, and direct technological support from the MEDCs.

Weakening the Inequality of Financial Support

Currently, the funding for NCD is around 1~2% of the total health Official Development Assistance (ODA), despite resulting in a majority of the global disease burden and 74% of all deaths globally according to WHO. International donors should increase the financial support to a level that is proportionate. Nations could also identify infrastructure gaps across various LEDCs and allocate funds more efficiently, especially for the regions that are in great need. Financing could be done through both domestic and international ways. Domestically, governments can raise allocated revenues through health taxes on tobacco, alcohol and sugar-sweetened drinks, while also including NCD services in UHC benefit packages and reduce excessive spending. Internationally, donors can raise the 1 to 2% share of health ODA going to NCDs, and use pooled and blended financing. These solutions should be linked to each country's own situations.

Increasing Public Awareness

Nations could create programs that promote the risk factors like the social and commercial determinants of health of chronic disease and educate the public about its risks. This can be based on the “4x4 framework” or the “5x5 framework” that is also about the risk factors of chronic diseases, however nations should change the specific details based on their income level and lifestyle.

Moreover, nations can also educate patients that use cross-border healthcare about the importance of maintaining their own medical documentation.

Major parties involved

People's Republic of China

Through its “Healthy China 2030” vision, China has prioritized the enhancement of healthcare quality and efficiency. In Shanghai, the Integrated Community Multimorbidity Care Model (ICMCM) was implemented to provide various care for people with hypertension and stroke by utilizing big data analytics and standardized service processes. This unified chronic disease management information system is primarily based on electronic health records, and it gathers fragmented data from various places such as clinics, secondary and tertiary hospitals. By storing these data in the integrated system, China aims to make preventive care and records of chronic disease patients to be easily accessible at different levels of facilities. Roughly 88% of all deaths in China currently are due to chronic and non-communicable diseases. Although smoking has shown a decline, alcohol consumption, obesity, and poor physical activity have all trended upwards. In China, the less affluent Northwestern regions surprisingly show higher rates of chronic disease, because of less care access and health literacy. China likely supports voluntary and state-led digital health standards and big-data systems, but could be cautious about cross-border data transfer and it strongly protects national data control.

United Kingdom of Great Britain and Northern Ireland

The United Kingdom, specifically England, operates the National Health Service (NHS), which is a tax-funded system that provides universal healthcare free when using. NHS utilizes Quality and Outcomes Framework (QOF), which is a financial incentive to encourage general practices to provide ongoing high-quality management for conditions like diabetes to standardize chronic disease management. The UK also currently faces significant challenges in regards to underinvestment and systemic efficiency. Therefore, a shift from reactive hospital-based care to proactive community-based services is prioritized. Moreover, around 15 million people in England alone currently live with chronic health conditions. Specifically, people living in the most deprived socioeconomic areas experience the start of various chronic conditions around 10 to 15 years earlier than affluent areas. The United Kingdom likely supports interoperability standards and prevention, while also focusing on primary and community care.

Russian Federation

From a hospital-centered and patient care focused system, Russia is moving towards a more integrated European-style system to combat chronic disease. The health system is funded through Obligatory Medical Insurance (OMI), though it continues to struggle with budget deficits and unequal resource distribution. Recent structural changes in Russia include a focus on early disease detection and a significant reduction in hospital bed capacity. In addition to this, compared to Western Europe and North America, Russia has significantly higher mortality rates from cardiovascular disease. Men especially have significant premature mortality than women, and the gap between male and female life expectancy and preventable chronic disease death is higher than other countries. This may be due to behavioral factors like heavy alcohol consumption, smoking, and low fruit and vegetable intake. Russia would likely emphasize the importance of sovereignty and non-interference programs.

Republic of France

France provides healthcare using the Social Health Insurance (SHI) system which is overseen by the French government. SHI has achieved universal coverage through the Universal Health Coverage Act (CMU). For chronic disease management, France utilizes the "Long-Term Disease" (ALD) procedure, which provides patients with standardized care protocols and funds for payments of specific chronic disease conditions and is managed by French National Authority for Health (HAS). In France, there are various problems such as current economic inequalities and its aging population that affects chronic conditions. However, 38% of French adults live with at least one chronic condition, which is around the average of EU nations. France would likely support integrated, people-centered care and shared standards through ALD protocols.

United States of America

The United States of America mainly operates a private and market-driven healthcare system that has both advanced technological innovation and significant insurance fragmentation with high insurance costs. Currently, chronic conditions affect roughly 60% to 75% of American adults, and it also accounts for $4.1 trillion in annual healthcare spending. The US utilizes the Health Insurance Portability and Accountability Act (HIPAA), which is a federal law that protects sensitive patient health information from being disclosed without consent. It also relies heavily on HL7 FHIR standards to enable consumer data access and data liquidity, while supporting the global adoption of the Chronic Care Model (CCM), which aims to standardize healthcare delivery through community

resources, health system organization, and more. The US would likely support private-sector innovation and market-driven interoperability, while also protecting patient's data privacy.

Federal Republic of Germany

Germany's healthcare system also utilizes Social Health Insurance (SHI), and it puts strong emphasis on self-governance through the Federal Joint Committee (GBA). Moreover, Germany introduced verified Disease Management Programmes (DMPs) for conditions such as type 2 diabetes, breast cancer, and coronary heart disease to improve care for the patients. This gives financial incentives for both sickness funds, health insurances provided by non-profit and self-governing public-law corporations, and doctors to register patients and maintain standardized documentation. In Germany, 53.7% of adults reported a chronic disease or long-term health problem, which is higher than most European nations, and the most prominent risk factor is nutrition-related, followed by high BMI and blood pressure. Germany would also support a unified care system and common standards through DMPs.

State of Japan

With nearly one in three citizens aged 65 or older, Japan has one of the world's largest proportion of aged population, placing great pressure on its universal healthcare system. The Japanese government has proposed an "Integrated Community Care System" to be established by 2025, which connects primary care, social services, and long-term care. Japan also uses a unique standardized model that combines medical checks with targeted health guidance for all insured individuals between the ages of 40 and 74. Japan currently has a high percentage of non-communicable chronic diseases, and it accounts for roughly 82% of all deaths, but has a higher expected total lifespan and lower obesity rate in comparison to other nations. Japan would likely support UHC-based approaches for its aging-population and community care solutions.

State of Israel

Israel has maintained its national Electronic Health Records (EHR) for over two decades, and is also a global pioneer in digital health transformation. Israel utilizes the "Eitan" platform, a health information exchange system that allows the seamless flow of clinical data between all healthcare organizations. Israel also adopted HL7 FHIR to ensure that its digital health ecosystem is interoperable and open to other nations. In Israel, there are mixed rates of chronic disease across

people's economic level and demographics. Overall, the nation has lower death rates but high rates of diabetes compared to other nations. Israel would likely promote FHIR-based exchange and digital cooperation.

Republic of South Africa

South Africa operates an Integrated Chronic Disease Management (ICDM) framework, which could serve as a critical model for middle-income nations. ICDM aims to optimally approach both communicable diseases and non-communicable diseases by providing integrated prevention and treatment at the primary healthcare level in order to provide a better outcome. In comparison to other nations, South Africa is uniquely facing both non-communicable diseases and communicable diseases. NCDs such as hypertension and diabetes, which is further affected by its high obesity rates, also affect South Africa, while other medical conditions such as HIV and AIDS also persist. South Africa likely supports integrated care and financing for LEDCs and middle-income countries, with attention to the double burden of communicable and non-communicable disease.

Republic of India

India represents a large, federated, and resource-diverse health system that has pioneered national digital public infrastructure through the Ayushman Bharat Digital Mission (ABDM) and the National Health Claims Exchange (NHCX). However, ecosystem-wide adoption remains constrained by paper-based legacy workflows in rural primary care and a lack of financial incentives. India would likely demand international support for technical capacity-building and affordable, scalable open-source solutions like digital goods.

Republic of Brazil

Brazil guarantees healthcare as a universal constitutional right through its Sistema Único de Saúde (SUS), which is one of the world's largest public health systems. While Brazil is facing challenges related to regional socioeconomic disparities and resource allocation, Brazil is also actively incorporating digital health policies and HL7 FHIR standards. Brazil would likely emphasize that health standardization must focus on social determinants of health, and primary healthcare (PHC) integration.

European Union (EU)

The European Union is a global leader in health data governance, with its basis on the General Data Protection Regulation (GDPR), which is a comprehensive European Union data privacy law. Following its formal adoption in 2024~2025, the EU officially published the European Health Data Space (EHDS) Regulation, which launched the world's first sectoral data space with multiple countries. Using the European EHR Exchange Format (EEHRxF), EHDS creates a harmonized legal and technical architecture. This enables seamless primary data exchange for cross-border care and regulated secondary data reuse for public health and scientific research. The EU would likely urge nations to adopt interoperable digital health spaces such as those modeled on the EHDS.

Timeline Of Events

Date	Description of Event
Framingham Heart Study, 1947	The United States National Heart Institute launches a landmark study to analyze the causes of chronic diseases by quantifying lifestyle risk factors.
Consultative Committee for Europe's Early NCD Interest, May 1951	The Consultative Committee for Europe recommends WHO to develop specific programs that targets non-infectious chronic diseases in adult populations.
Amsterdam Symposium, October 1957	Global health officials group cardiovascular disease, diabetes, respiratory illnesses, and cancer as degenerative diseases linked to lifestyle behaviors.
North Karelia Project, 1978	Finland initiates a community-based intervention project that serves as a template for reducing lifestyle risk factors to mitigate NCD rates.
Integrated NCD Approach, 1982	WHO Expert Committee concludes that efforts against heart disease should be part of a general program against NCDs due to shared risk factors.
Launch of CINDI, 1984	The WHO Regional Office for Europe formally invites countries to join the Countrywide Integrated Noncommunicable Diseases Intervention program. (CINDI)
Launch of INTERHEALTH, 1990	The WHO integrates community-based approaches across all regions to combat the common risk factors for four major NCDs.
Introduction of NEHR, 1992	The earliest National Electronic Health Record (NEHR) system was introduced in the WHO European region.

Creation of the 4x4 Framework, May 2000	The 53rd World Health Assembly adopts resolution WHA53.17, standardizing the global focus on four major diseases and four shared behavioral risk factors.
Global NCD Action Plan, 2008	The WHO releases the Action Plan for the Global Strategy, prominently featuring the 4x4 model as the core of NCD policy.
Launch of Global HEARTS, 2016	The WHO launches an initiative to support countries in strengthening cardiovascular disease prevention through primary healthcare.
WHA Resolution on Digital Health, May 2018	Resolution WHA71.7 requests a global strategy on digital health to identify priority areas for WHO efforts.
Shift to the 5x5 Agenda, September 2018	The third UN High-Level Meeting on NCDs broadens the global framework from the traditional 4x4 to a 5x5 agenda. It officially expands commitments to include mental health and environmental risk factors.
Digital Health Strategy Endorsement, 2020	The Seventy-third World Health Assembly endorses the Global Strategy on Digital Health 2020–2025.
Record NCD Mortality, 2021	Noncommunicable diseases are reported to have killed at least 43 million people, accounting for 74% of global non-pandemic deaths.
WHO EURO Digital Health Survey, 2022	A comprehensive assessment evaluates the status of electronic health records and data exchange across 53 Member States.

UN Involvement, Resolutions, Treaties and Events

- WHO Global Strategy on Digital Health (2020–2025)

- Endorsed by the Seventy-third World Health Assembly in 2020, this strategy provides a comprehensive roadmap to strengthen health systems through the application of digital technologies. Promoting global collaboration, advancing the implementation of national digital health strategies, strengthening global governance, and advocating for people-centered health systems are the objectives of this. The strategy aims to bridge the "digital divide" by prioritizing digital public goods and establishing global minimum standards for EHRs.

- SDGs Target 3.4 (The 2030 Agenda for Sustainable Development)

- Sustainable Development Goals acknowledges chronic diseases as a primary driver of poverty and a significant obstacle to global economic growth. In SDG Target 3.4, it mentions that all Member States have committed to reducing premature mortality from NCDs by one-third by the year 2030 by enhancing various prevention and treatment efforts. WHO also called for the development of an Implementation Roadmap to accelerate progress on nine voluntary global targets.

- Fourth High-Level Meeting (HLM4) on Noncommunicable Diseases and Mental Health (September 2025)

- To accelerate stalled progress toward Sustainable Development Goal (SDG) target 3.4 mentioned above, the declaration established global benchmarks to reach by 2030. These include achieving 150 million fewer tobacco users worldwide and bringing 150 million more people’s high blood pressure under control. In addition to this, the declaration also binds member states to reach various health system metrics to meet by 2030. 80% of nations must implement operational multisectoral plans and regulatory, legislative, or fiscal health policies, and 80% of primary health care facilities globally must be reliably stocked with basic technologies and essential medicines for NCDs and mental health.

- Resolution WHA53.17 (May 2000)

- This resolution formally adopted the "Global Strategy for the Prevention and Control of Noncommunicable Diseases,", which meant the policy creation of NCD management at the WHO. It established the pervasive "4x4 framework," focusing on 4 diseases—cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes—and 4 behavioral risk factors—tobacco use, harmful use of alcohol, unhealthy diets, and physical inactivity. This framework served as a simplified governance tool for the WHO, supporting emerging NCD policies and standardizing global monitoring.

- WHO Framework Convention on Tobacco Control (WHO FCTC, 2003): The WHO's first international legal treaty that negotiates global rules on tobacco price, tax measures, advertising, and packaging. This framework demonstrates how legally binding standardization can effectively curb a primary NCD risk factor.

Bibliography

"About Chronic Diseases | Chronic Disease." *CDC*, 2026, Accessed Aug 1 2026, https://www.cdc.gov/chronic-disease/about/index.html.

Alexandra Dimmer, Robert Baird, Pramod Puligandla. "Role of practice standardization in outcome optimization for CDH." *PMC*, 2024, Accessed Aug 1 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10961560/.

"Cardiovascular diseases (CVDs)." *World Health Organization (WHO)*, 31 July 2025, Accessed Aug 1 2026, https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds).

"Chronic diseases." *OECD*, Accessed Aug 1 2026, https://www.oecd.org/en/topics/policy-issues/chronic-diseases.html.

"Chronic respiratory diseases." *World Health Organization (WHO)*, Accessed Aug 1 2026, https://www.who.int/health-topics/chronic-respiratory-diseases#tab=tab_1.

"COPD - What Is COPD? | NHLBI, NIH." *National Heart, Lung, and Blood Institute (NHLBI)*, 2024, Accessed Aug 1 2026, https://www.nhlbi.nih.gov/health/copd.

"EHR Barrier: Standards (Lack of Interoperability Standards)." *World Health Organization | European Region*, 2026, Accessed Aug 1 2026, https://gateway.euro.who.int/en/indicators/hisgov_61-ehr-barrier-standards-lack-of-interoperability-standards/#id=38446.

"Electronic Health Records." *CMS*, 2024, Accessed Aug 1 2026, https://www.cms.gov/priorities/key-initiatives/e-health/records.

"Electronic health records and data exchange in the WHO European region: A subregional analysis of achievements, challenges, and prospects." *ScienceDirect*, 2025, Accessed Aug 1 2026, https://www.sciencedirect.com/science/article/pii/S1386505624003502.

Ellen Nolte, Cécile Knai, Martin McKee. "Managing chronic conditions Experience in eight countries." *European Observatory on Health Systems and Policies*, 2008, Accessed Aug 1 2026, https://iris.who.int/server/api/core/bitstreams/67fba441-bc88-4cc8-a62b-e14f8892bff3/content .

Esti Shelly. "Digital Health Transformation: The Israeli Experience." *State of Israel Ministry of Health*, 2020, Accessed Aug 1 2026, https://www.medizininformatik-initiative.de/sites/default/files/2020-07/2020-02-13_Esti%20Shelly_Digital%20Health%20Transformation.pdf.

"Global strategy on digital health 2020-2025." *World Health Organization*, 2021, Accessed Aug 1 2026, https://www.who.int/docs/default-source/documents/gs4dhdaa2a9f352b0445bafbc79ca799dce4d.pdf.

Hanson Mark, Gluckman Peter. "Developmental origins of noncommunicable disease: population and public health implications." *ScienceDirect.com*, 2011, Accessed Aug 1 2026, https://www.sciencedirect.com/science/article/pii/S0002916523025868.

Hatefi, Arian. “Global susceptibility and response to noncommunicable diseases.” *PMC*, 2018, Accessed Aug 1 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6083392/.

“Healthcare’s standardization solution.” *Kenan-Flagler Business School*, 2025, Accessed Aug 1 2026, https://www.kenan-flagler.unc.edu/news/healthcares-standardization-solution/.

“INTEROPERABILITY | English meaning - Cambridge Dictionary.” *Cambridge Dictionary*, Accessed Aug 1 2026, https://dictionary.cambridge.org/dictionary/english/interoperability.

Krugman, Allison. “Around the World, Chronic Diseases Are Rising.” *Think Global Health*, 2025, Accessed Aug 1 2026, https://www.thinkglobalhealth.org/article/around-the-world-chronic-diseases-are-rising.

Leah N. Schwartz, Jonathan D. Shaffer, Gene Bukhman. “The origins of the 4 × 4 framework for noncommunicable disease at the World Health Organization.” *Sciencedirect.com*, 2021, Accessed Aug 1 2026, https://www.sciencedirect.com/science/article/pii/S2352827321000069.

Meirmanov Serik. “Current healthcare system in Japan: Current issues and future directions.” *EHDI*, Accessed Aug 1 2026, https://www.journalehdi.com/download/current-healthcare-system-in-japan-current-issues-and-future-directions-17377.pdf.

Mengyun Sui. “The digitized chronic disease management model: scalable strategies for implementing standardized healthcare and big data analytics in Shanghai.” *frontiers*, 2023, Accessed Aug 1 2026, https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2023.1241296/full.

"The Missing Link in Cross-Border Healthcare: Why Medical Records Matter More Than You Think." *Kogod School of Business*, 2026, Accessed Aug 1 2026, https://kogod.american.edu/news/the-missing-link-in-cross-border-healthcare-why-medical-records-matter-more-than-you-think.

Mustafa Mahir Ulgu. "A Nationwide Chronic Disease Management Solution via Clinical Decision Support Services: Software Development and Real-Life Implementation Report." *JMIR Publications*, 2024, Accessed Aug 1 2026, https://medinform.jmir.org/2024/1/e49986.

Nicolas Xu. "A Comprehensive Review of Multinational Healthcare System Performances: United States, United Kingdom, China, Russia, Tanzania, Australia, and Brazil." *Research Archive of Rising Scholars*, 2025, Accessed Aug 1 2026, https://research-archive.org/index.php/rars/preprint/view/2345.

"Noncommunicable diseases." *World Health Organization (WHO)*, 25 September 2025, Accessed Aug 1 2026, https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases.

Pratima Bhuttarowas, PhD, RN. "The Importance of Care Standardization in Medical Practice." *Zynxhealth*, Accessed Aug 1 2026, https://www.zynxhealth.com/insights/importance_of_care_standardization/.

Shishkin, Sergey. "Structural changes in the Russian health care system: Do they match European trends?" *Econstar*, 2022, Accessed Aug 1 2026, https://www.econstor.eu/bitstream/10419/285266/1/1813305668.pdf.

Sinsky, Christine A. "Standardization vs Customization: Finding the Right Balance." *PMC*, 2021, Accessed Aug 1 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7939702/.

"What Is a Chronic Illness? How To Cope." *Cleveland Clinic*, 2025, Accessed Aug 1 2026, https://my.clevelandclinic.org/health/articles/4062-chronic-illness.